C and C++

Duration: 3-6 Months | Mode: HYBRID | 100% Placement Support

Detailed Curriculum & Module Outline:

Module 1: C Programming Basics

Compilation, data types, operators, control flow and functions, run and inspected rather than j

Module 2: Arrays, Strings and Functions

How arrays sit in memory, string handling in C, scope, recursion and parameter passing

Module 3: Pointers and Memory Management

Address arithmetic, malloc and free, dangling pointers and leaks - the module the rest of the c

Module 4: Structures, Unions and File Handling

Composite types, reading and writing files, and modelling records without a database

Module 5: C++ and Object Oriented Basics

Classes, objects, constructors, destructors and the differences from C that actually matter

Module 6: Inheritance and Polymorphism

Base and derived classes, virtual functions, abstract classes and when inheritance is the wrong

Module 7: Operator Overloading and Templates

Writing generic, reusable code, and the cost of getting an interface subtly wrong

Module 8: The Standard Template Library

Vectors, maps, sets, iterators and algorithms - and choosing the right container for the job

Module 9: Exception Handling and File Streams

Errors that surface instead of silently corrupting state, plus stream-based input and output

Module 10: Debugging and Problem Solving

GDB, breakpoints, watchpoints and reading a crash, applied to real broken programs

Module 11: Mini Projects

Two console applications built end to end, reviewed for structure and memory behaviour as well